

You Look Only Once (YOLOv1) for Object Detection

Introduction to YOLOv1

You Only Look Once (YOLO) proposes using an end-to-end neural network that makes predictions of bounding boxes and class probabilities all at once.

Introduced by Joseph Redmon et al. in 2015 in the paper “**You Only Look Once: Unified, Real-Time Object Detection**”

Ref:

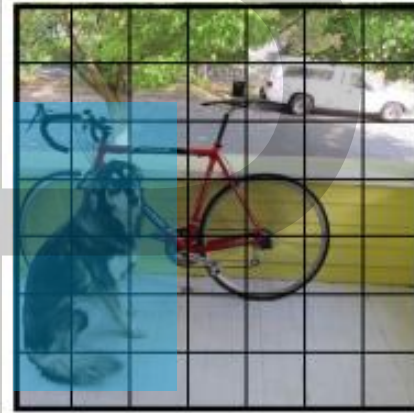
Redmon, J., Divvala, S., & Farhadi, A. (2015). You Only Look Once: Unified, Real-Time Object Detection. *arXiv preprint arXiv:1506.02640*. [arXiv](#)

How YOLOv1 works

Input Image



Each grid cell is responsible for detecting objects whose center falls inside that cell.

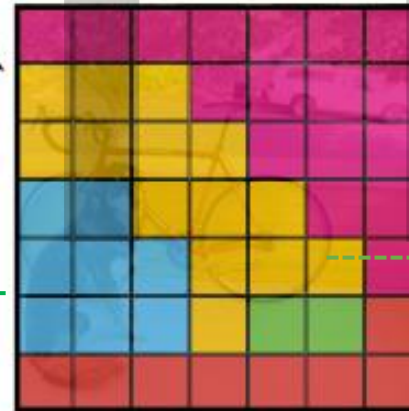


$S \times S$ grid on input



Bounding boxes + confidence

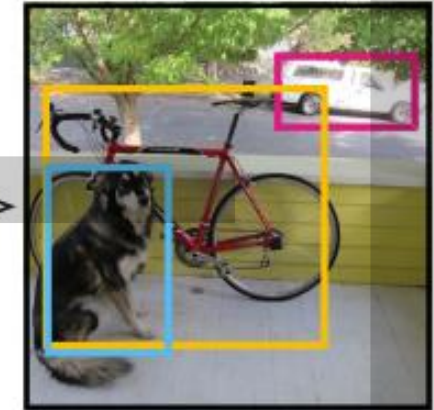
Each grid cell outputs
✓ Bounding boxes
(position + size)
✓ Confidence scores



Class probability map

Dog Class

Cycle class



Final detections



Why so many rectangles?



Each grid cell predicts **multiple bounding boxes**.

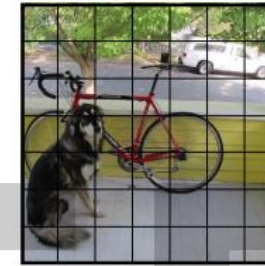
Each predicted box has:

- ✓ **Position (x, y)**
Center relative to the grid cell.
- ✓ **Size (w, h)**
Relative to the whole image.



Also predicts Confidence score

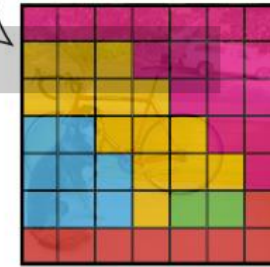
$$\text{Probability (object exists)} \times IoU_{pred}^{truth}$$



5 x 5 grid on input



Bounding boxes + confidence



Class probability map



Final detections

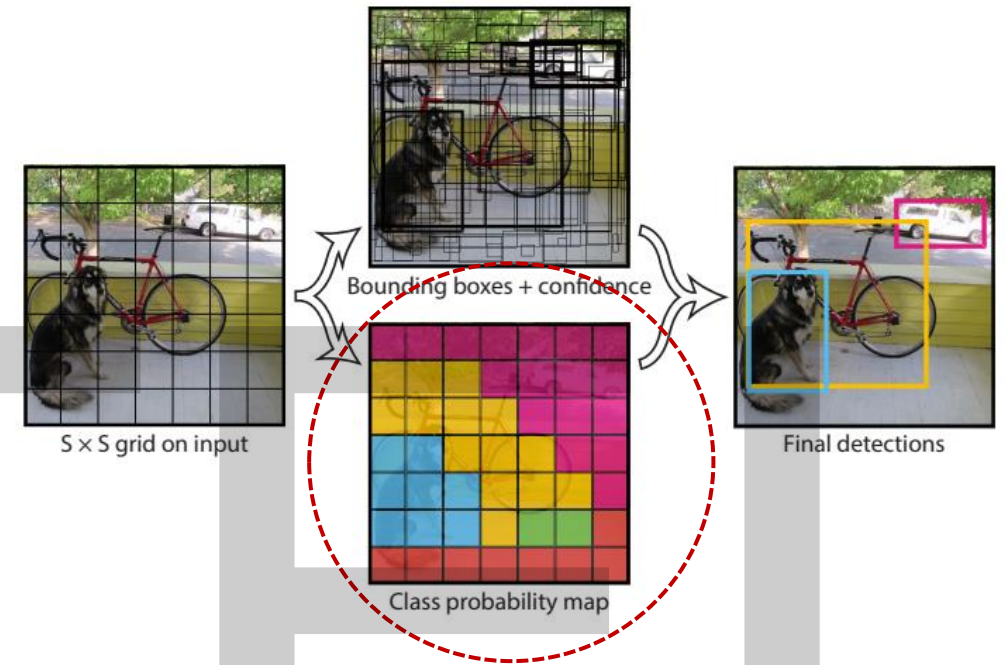


dog: 0.90
bicycle: 0.06
car: 0.04

- ✓ Each grid predicts conditional class probability $Pr(Class_i | Object)$
Ex: $Pr(dog \text{ class} | Object \text{ exists})$

During Test Time

- ✓ One set of class probabilities per grid cell, regardless of the number of bounding boxes



Confidence score for the bounding box

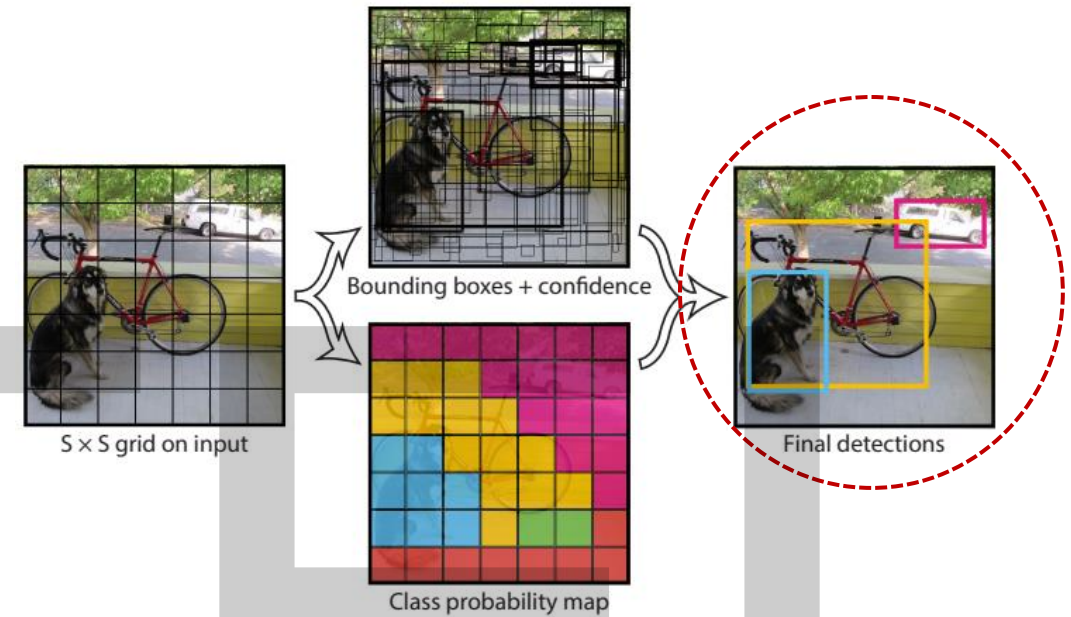
$$Pr(Class_i | Object) * Pr(Object) * IoU_{pred}^{truth} = Pr(Class_i) * IoU_{pred}^{truth}$$

- ✓ Gives class-specific confidence scores for each box.

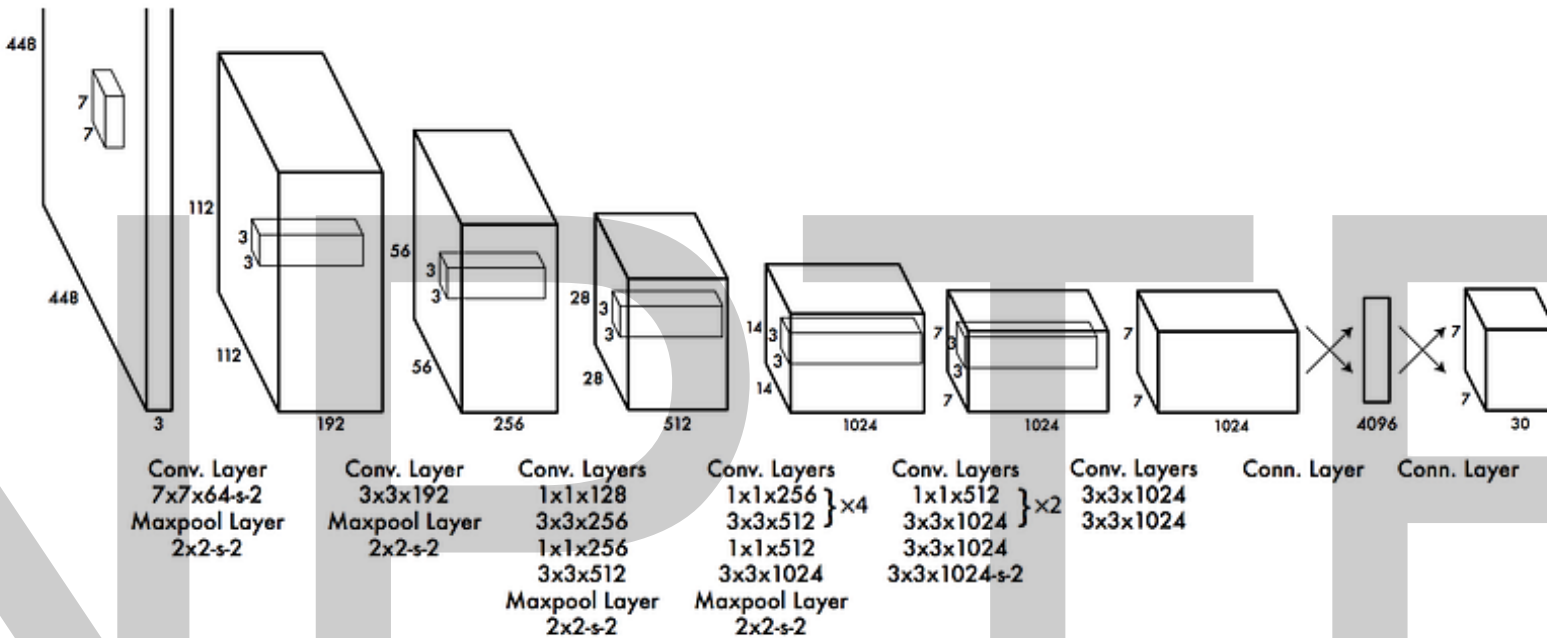


Non-Max Supression

- ✓ An object can have multiple boxes with IOU beyond the threshold,
- ✓ Keeping all those boxes might include noise.
- ✓ Non-Max Suppression to keep only the boxes with the highest probability detection score.
- ✓ Non-Max suppression is used during inference



Architecture of YOLOv1 Model



- It has 24 convolutional layers followed by 2 fully connected layers.
- Initial convolutional layers of the network extract feature from the image
- Fully connected layers predict the output probabilities and coordinates.

YOLOv1 treats detection as a regression problem.

For each grid cell it predicts:

- ✓ Bounding box coordinates: x, y, w, h
- ✓ Confidence score: C
- ✓ Class probabilities: $p(class_i)$

Loss function in YOLOv1

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$

This term is 1 **only** for the box responsible for detecting the object.

Localization Loss
(position + size)

Penalizes incorrect prediction of x,y,
Default value: 5

$$+ \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right]$$

Penalizes bad prediction of box sizes.

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2$$

Ground truth confidence = 1

Confidence Loss
(object vs no object)

$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2$$

Ground truth confidence = 0

Default: 0.5

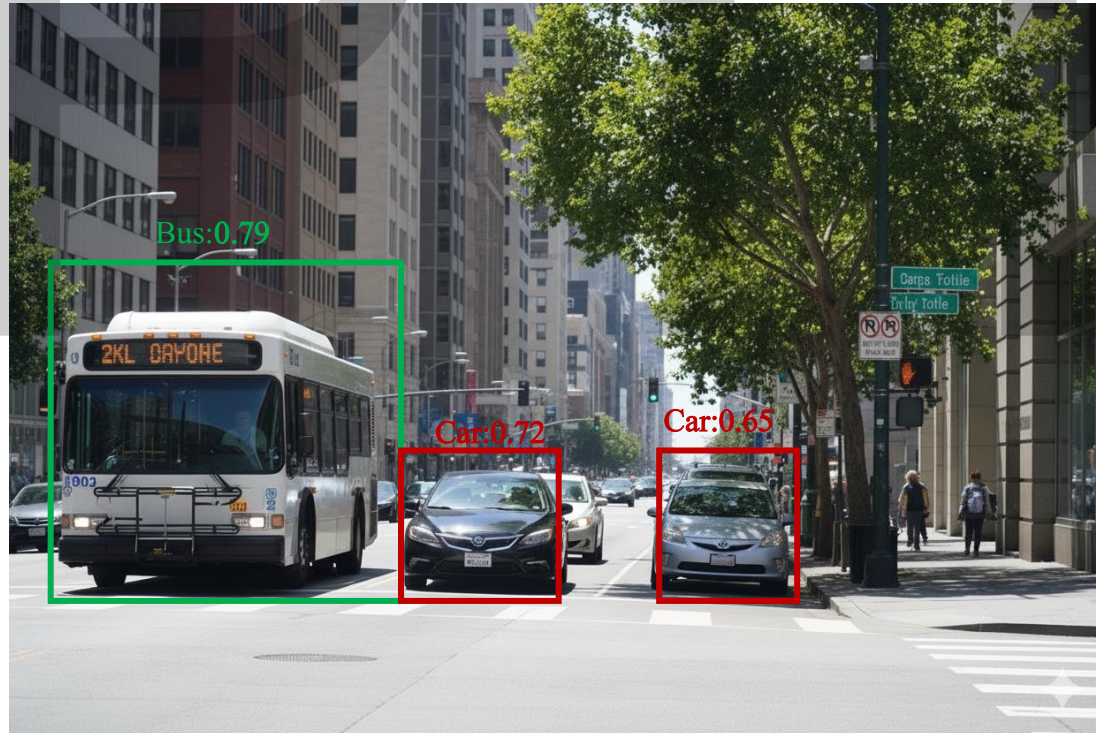
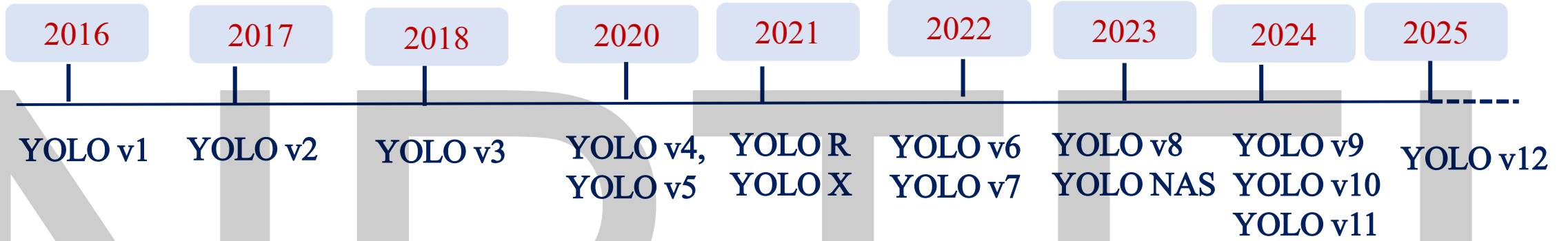
This term is 1 **only** for the box without the object.

Penalizes incorrect class probability predictions.

$$+ \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2$$

Classification loss
(per class)

Variants of YOLO



NPTEL

Summary

Next Session

Hands-on session on U-Net Architecture

AND

Object detection using YOLO and

NPTEL

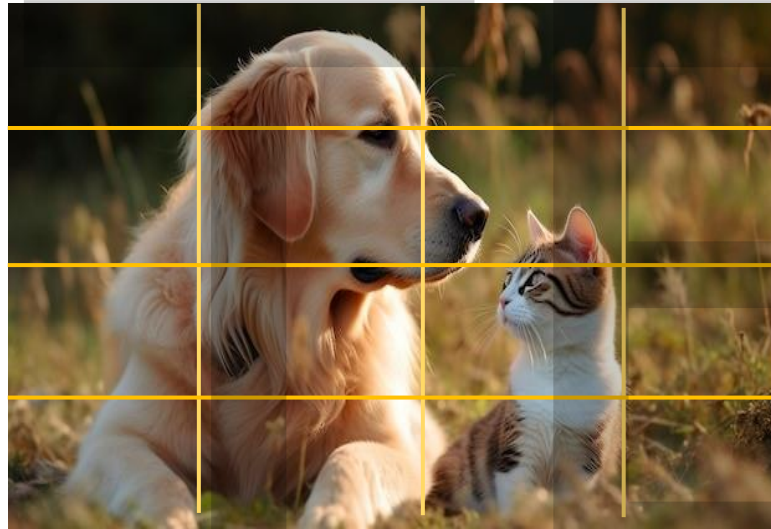
Thank you

Object Detection by YOLO

1. Residual blocks



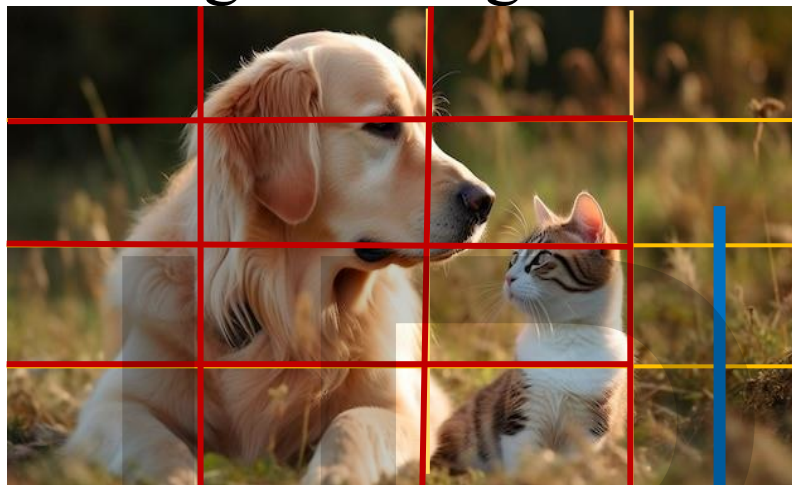
Input Image



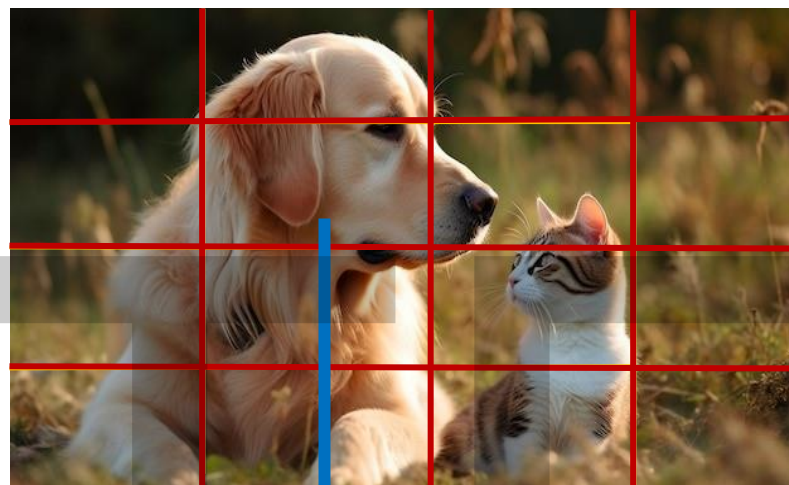
4 x 4 Grid

Each cell in the grid is responsible for localizing and predicting the class of the object that it covers, along with the probability/confidence value.

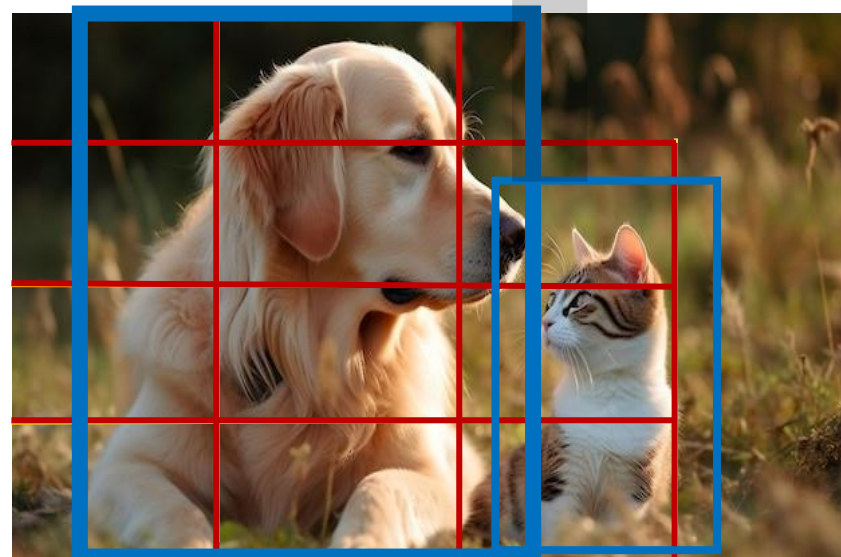
2. Bounding Box Regression

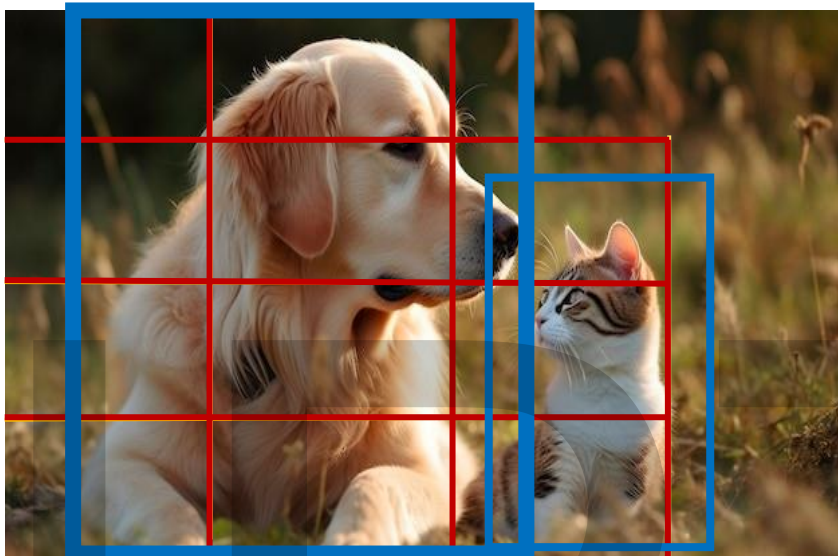


Grids with no object



Significant grids





Output $Y = [pc, \underbrace{bx, by}_{x, y \text{ coordinates of the center of the bounding box w.r.t the enveloping grid cell}}, \underbrace{bh, bw}_{\text{height and width of the bounding box w.r.t the enveloping grid cell.}}, \underline{c1, c2}]$

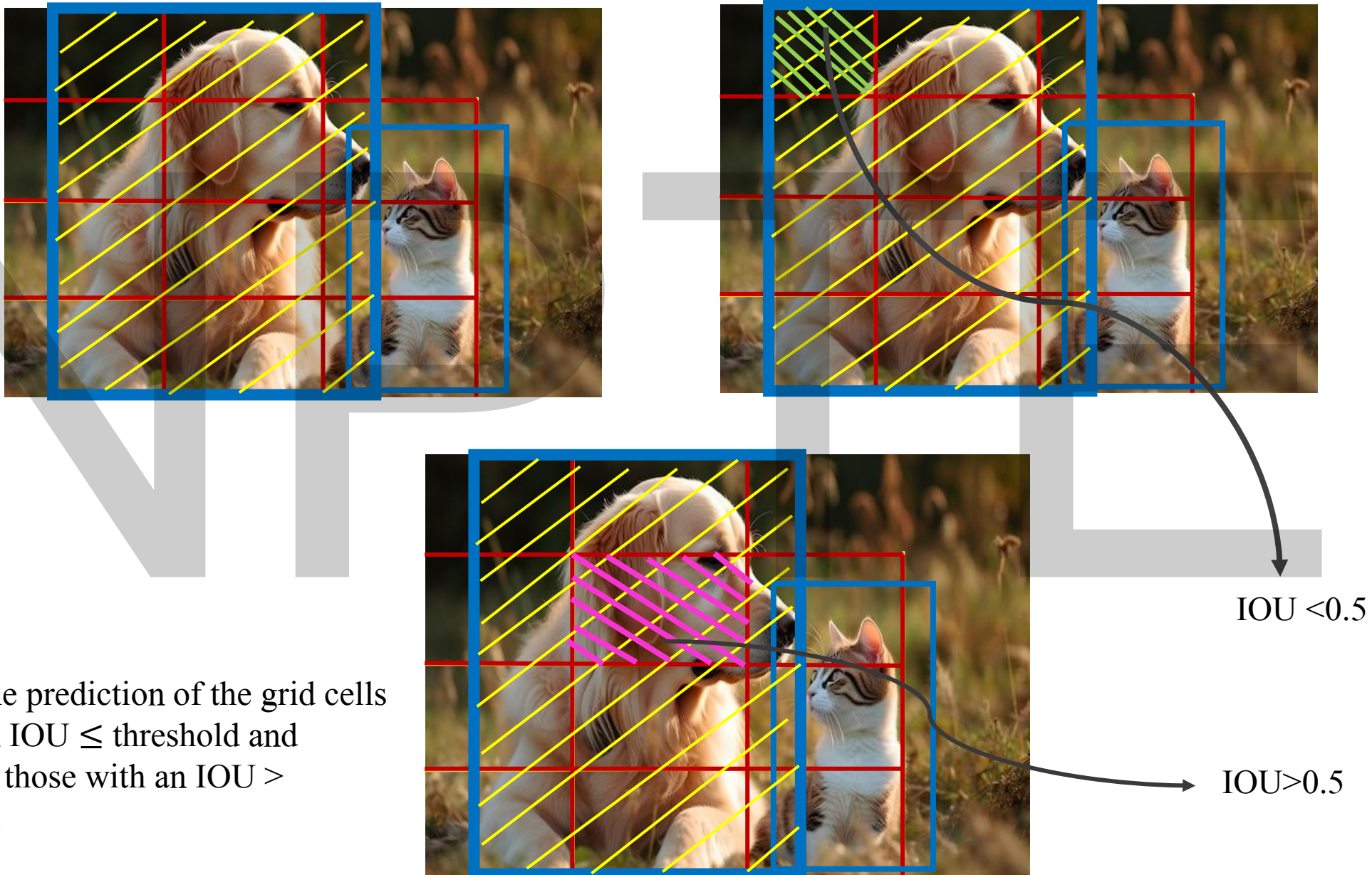
Probability score of the grid containing an object

x, y coordinates of the center of the bounding box w.r.t the enveloping grid cell.

height and width of the bounding box w.r.t the enveloping grid cell.

correspond to the two classes, cat and dog.

3. Intersection Over Unions or IOU



ignores the prediction of the grid cells having an $\text{IOU} \leq \text{threshold}$ and considers those with an $\text{IOU} > \text{threshold}$.

YOLOV1 and YOLOV2

	YOLOV1	YOLOV2	Benefit
Bounding Box prediction	Predicted bounding boxes directly from cell coordinates.	Adopted anchor boxes (like Faster R-CNN), which lets the model predict multiple boxes per grid cell with better aspect ratios.	More accurate localization, can detect multiple objects in one cell.
Input Resolution	Trained on 224×224 input.	Increased to 448×448 input resolution for detection.	Captures finer details and improves detection of small objects.
Backbone architecture	Custom 24-layer conv net	Darknet-19 backbone: 19 conv layers + 5 max-pool layers.	Faster & more accurate feature extractor
mAP (on VOC 2007)	63.4	76.8	

YOLOV2 and YOLOV3

	YOLOV2	YOLOV3	Benefit
Bounding Box prediction	Anchor boxes (multiple boxes per cell)	Anchor boxes at 3 different scales (multi-scale feature maps)	Detects small, medium, large objects more reliably
Detection	Single scale detection	Multiscale detection	Predicts larger objects in initial layers and smaller objects in final layers
Backbone architecture	Darknet-19 backbone: 19 conv layers + 5 max-pool layers.	Darknet-53 (53 conv layers with residual connections)	Deeper & stronger features → higher accuracy
mAP (on VOC 2007)	76.8	80+	
Speed	~67 FPS	~30 FPS	Accuracy improved at some cost in speed